PID Controller – Tutorial 1

$$G_1(s) = \frac{(s+2)}{(s+1)(s-3)(s+4)}$$ unstable

Zero: $(s+2) \Rightarrow 0 \Rightarrow \underline{s = -2}$

Poles: $(s+1)(s-3)(s+4) \Rightarrow 0 \Rightarrow$

$s = -1$ left half
$\times\ s = 3$ right half $\times$
$s = -4$ left half

since it has one pole in the right ~~s~~ side of the vertical axis, it is unstable.

$$G_2(s) = \frac{1}{s^2 + s + 1}$$ stable

Zero: no zero

Poles: $\left(s + 0.5 - j\frac{\sqrt{3}}{2}\right)\left(s + 0.5 + j\frac{\sqrt{3}}{2}\right) \Rightarrow 0$

$\checkmark\ s = -0.5 + 0.866j$
$\checkmark\ s = -0.5 - 0.866j$

stable

$$G_3(s) = \frac{s+1}{s^2 - s + 1}$$ unstable

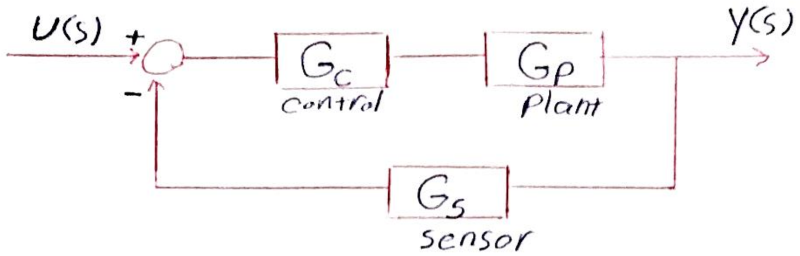
Zero: $s = -1$

Poles: $\left(s - \frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\left(s - \frac{1}{2} - j\frac{\sqrt{3}}{2}\right)$

$\times\ s = \frac{1}{2} + j\frac{\sqrt{3}}{2}$
$\times\ s = \frac{1}{2} - j\frac{\sqrt{3}}{2}$

unstable

2)



Close loop Transfer function

$$G(s) = \frac{Y(s)}{U(s)} = \frac{G_c^{(s)} G_p^{(s)}}{1 + G_c^{(s)} G_p^{(s)} G_s^{(s)}}$$

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a)

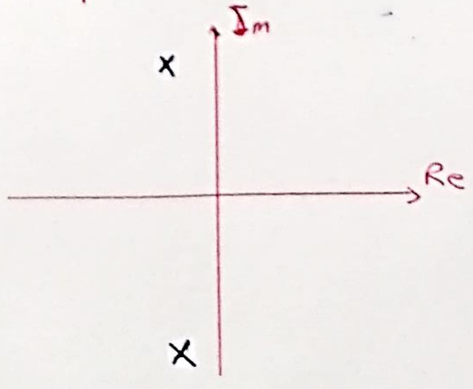
$$G_c = 10 \qquad G_p = \frac{2}{s(s+1)} = \frac{2}{s^2+s} \qquad G_s = 1$$

$$\Rightarrow G(s) = \frac{\frac{20}{s^2+s}}{1 + \frac{20}{s^2+s}} = \frac{\frac{20}{s^2+s}}{\frac{s^2+s+20}{s^2+s}} = \frac{20}{s^2+s+20}$$

$$\Rightarrow G(s) = \frac{20}{s^2+s+20} = \frac{20}{\left(s+\frac{1}{2} \pm j\frac{\sqrt{79}}{2}\right)}$$

Poles:

$$\left.\begin{aligned} s_1 &= -\frac{1}{2} + j\frac{\sqrt{79}}{2} \\ s_2 &= -\frac{1}{2} - j\frac{\sqrt{79}}{2} \end{aligned}\right\} \Rightarrow \text{Stable}$$



2 b)

$G_c(s) = 5 + \frac{3}{s}$ $\quad G_p(s) = \frac{2}{s(s+1)}$ $\quad G_s(s) = 1$

$$G(s) = \frac{G_c(s)G_p(s)}{1 + G_c(s)G_p(s)G_s(s)} = \frac{\left(\frac{5s+3}{s}\right)\left(\frac{2}{s(s+1)}\right)}{1 + \left(\frac{5s+3}{s} \times \frac{2}{s(s+1)}\right) \times 1}$$

$$G(s) = \frac{\frac{10s+6}{s^2(s+1)}}{1 + \frac{10s+6}{s^2(s+1)}} = \frac{\frac{10s+6}{\cancel{s^2(s+1)}}}{\frac{s^3+s^2+10s+6}{\cancel{s^2(s+1)}}} = \frac{10s+6}{s^3+s^2+10s+6}$$

characteristic equation (polynomial)

$$s^3 + s^2 + 10s + 6 = 0$$

Routh-Hurwitz

$$\begin{array}{c|cc} s^3 & 1 & 10 \\ s^2 & 1 & 6 \\ s^1 & \left(\frac{10-6}{1}\right) = 4 & 0 \\ s^0 & \frac{4 \times 6 - 0}{4} = 6 & \end{array}$$

No change in sign $\longrightarrow$ system stable